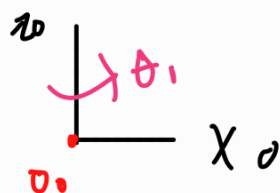
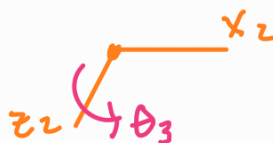
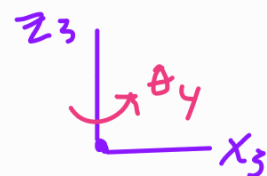
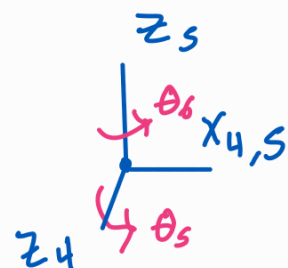
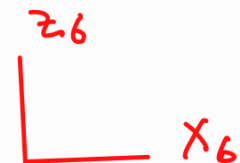


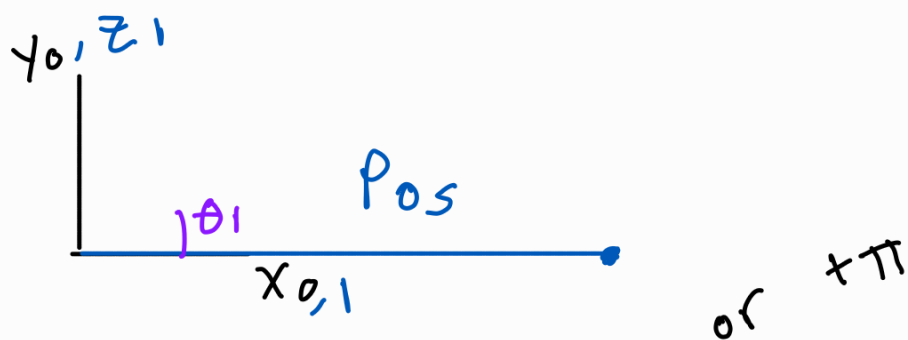
Link	a	d	α	θ
1	0	d_1	$\pi/2$	θ
2	a_2	0	0	θ
3	a_3	0	$-\pi/2$	θ
4	0	d_4	$\pi/2$	θ
5	0	0	$-\pi/2$	θ
6	0	d_6	0	θ

a)

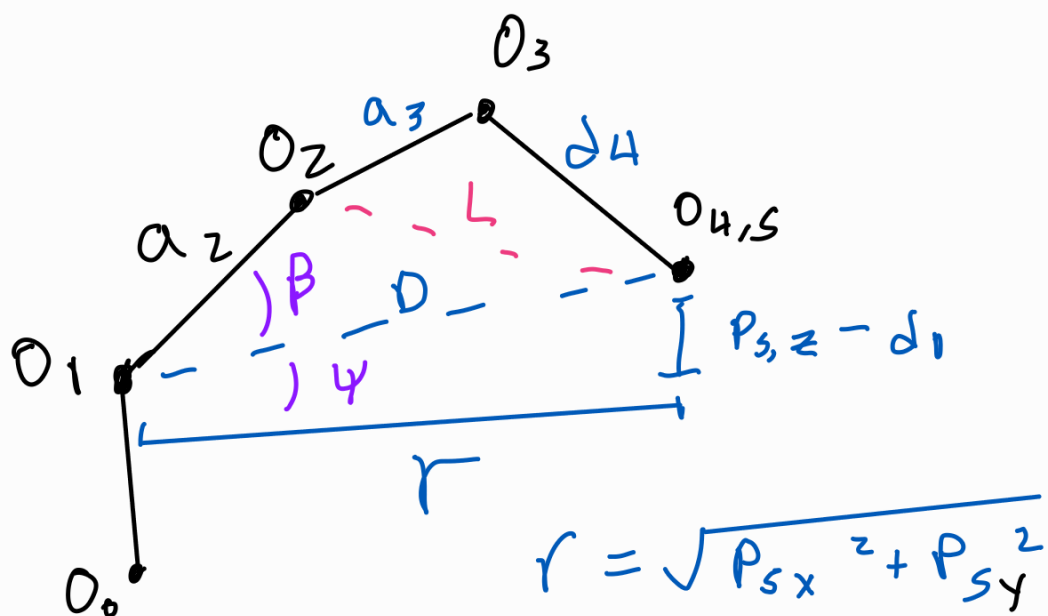


b) Let $P_{0,6} = [P_{6x}, P_{6y}, P_{6z}]^T$

$$P_{05} = P_{06} - d_6 \cdot R(:, 3)$$



$$\therefore \theta_1 = \text{atan}(P_{05,y}, P_{05,x})$$



$$\gamma = \text{atan2}(P_{5,z} - d_1, \sqrt{P_{5x}^2 + P_{5y}^2})$$

$$C^2 + S^2 = 1$$

$$\theta_2 = \psi \pm \beta$$

$$L^2 = a_2^2 + D^2 - 2a_2 D C(\beta)$$

$$D = \sqrt{\sqrt{P_{sx}^2 + P_{sy}^2} + (P_{sz} - d_1)^2}$$

$$L = \sqrt{a_3^2 + d_4^2}$$

$\therefore$

$$C(\beta) = \frac{a_2^2 + D^2 - L^2}{2a_2 D}$$

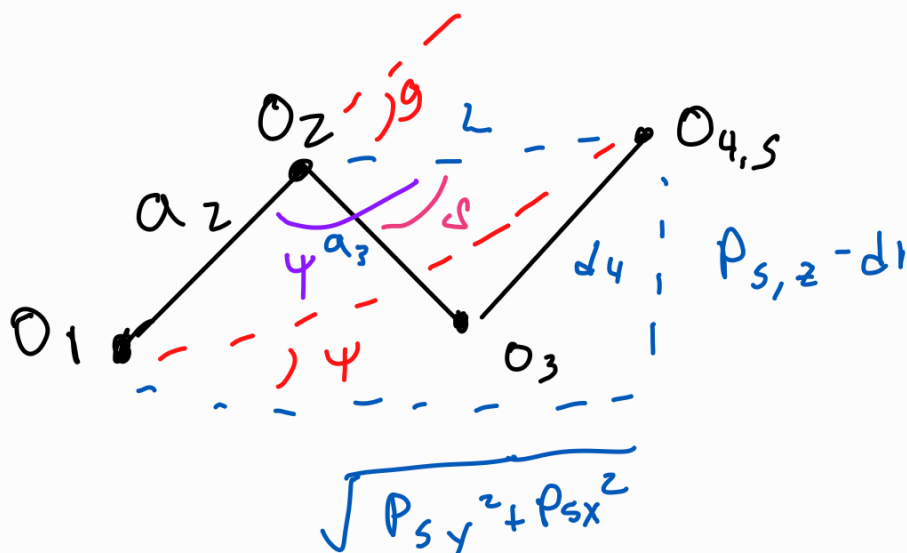
$$S(\beta) = \pm \sqrt{1 - C^2(\beta)}$$

$$\beta = \arctan 2 \left(\pm \sqrt{1 - \frac{L^2 - a_2^2 - D^2}{2a_2 D}}, \frac{L^2 - a_2^2 - D^2}{2a_2 D} \right)$$

$$\theta_z =$$

$$\arctan 2 \left(P_{s,z} - d_1, \sqrt{P_{sx}^2 + P_{sy}^2} \right)$$

$$+ \arctan 2 \left(\sqrt{1 - \left(\frac{a_z^2 + D^2 - L^2}{2a_z D} \right)^2}, \frac{a_z^2 + D^2 - L^2}{2a_z D} \right)$$



$$\delta = \arctan 2 (d_4, a_3)$$

$$\theta = \pi - \gamma$$

$$\theta_3 = \theta - \delta = (\pi - \gamma) - \delta$$

$$D^2 = a_2^2 + L^2 - 2a_2L \cos(\gamma)$$

$$\therefore \cos \gamma = \frac{a_2^2 + L^2 - D^2}{2a_2L}$$

$$\sin \gamma = \sqrt{1 - \cos^2 \gamma}$$

$$\gamma = \arccos\left(\frac{a_2^2 + L^2 - D^2}{2a_2L}\right)$$

$\therefore$

$$\theta_3 =$$

$$\pi - \arccos\left(\sqrt{1 - \frac{a_2^2 + L^2 - D^2}{2a_2L}}, \frac{a_2^2 + L^2 - D^2}{2a_2L}\right)$$

$$- \arccos(d_4, d_3)$$

Find 0R_3 from DH table

$${}^3R_6 = {}^0R_3^{-1} \cdot {}^0R_6$$

This is $z \ y \ z$

$\therefore$

$$\theta_5 = \text{atan2} \left(\pm \sqrt{r_{13}^2 + r_{23}^2}, r_{33} \right)$$

$$\theta_4 = \text{atan2} \left(\frac{-r_{23}}{s\theta_5}, \frac{-r_{13}}{s\theta_5} \right)$$

$$\theta_6 = \text{atan2} \left(\frac{-r_{32}}{s\theta_5}, \frac{r_{31}}{s\theta_5} \right)$$